

Joshua Dinn

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Education

Monash University, *Melbourne, Australia*

Mar 2023 – Oct 2027

Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

Experience

Collingwood Football Club, *Data & Analytics Intern*

Mar 2026 – Jun 2026

- Built a membership churn prediction model in XGBoost on historical transaction and engagement data to flag at-risk members.
- Created an interactive dashboard breaking down churn by segment, using SHAP analysis to surface the behaviours driving retention.
- Set up a synchronisation workflow between the club's web and mobile app repositories to keep deployments consistent across teams.

Monash eSolutions, *AI Fitness Trainer*

Jul 2026 – Present

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Assist in delivering a university webinar on AI to an audience of 200+ attendees.

Monash Deep Neuron, *Optimised Computing Member*

Sep 2025 – Present

- Building the on-device AI memory system for AR glasses: a vector database with HNSW indexing and approximate nearest-neighbour (ANN) search powering a retrieval-augmented generation (RAG) pipeline that grounds responses in prior context with low latency.
- Teaching Assistant for an 8-week C++ and Accelerated Computing workshop on GPU programming, run with NVIDIA, Pawsey Supercomputing Centre, and NCI.
- Earned NVIDIA CUDA certification through the program.

Outlier AI, *AI Trainer (Mathematics)*

Jun 2024 – Dec 2025

- Reviewed and corrected AI-generated mathematical solutions written in LaTeX to improve accuracy and reasoning.
- Wrote detailed feedback on computational and logical errors across advanced problem sets.

TMJ Tutoring, *Mathematics Tutor*

Feb 2023 – Present

- Tutor high school students in mathematics, focusing on exam preparation and problem solving.

Projects

Algorithmic Trading System for S&P 500 ETF

May 2025

- Built and backtested a trading system using Hidden Markov Models (HMMs) to classify market regimes for SPY.
- Used RSI, ATR, and volatility signals to drive position sizing and trade execution.
- Deployed it on an AWS Linux server with the Alpaca API for automated paper trading.

Volatility Forecasting with a GARCH Model

Apr 2025

- Built a volatility forecasting pipeline for the S&P 500 using a GARCH(1,1) model with residual diagnostics and time-series testing, automating data processing, evaluation, and reporting in Python.

Interactive Maze Solver & Algorithm Visualiser

Oct 2025

- Built a pathfinding platform visualising 7 algorithms across procedurally generated mazes.
- Added automated benchmarking and metrics tracking across 125,000+ simulated maze runs.

Skills & Interests

Programming: Python, C++, JavaScript, Java, MATLAB, SQL

Data & Quantitative: pandas, NumPy, XGBoost, Snowflake, Excel, OxMetrics

Interests: Statistical modelling, financial econometrics, machine learning, data analysis